Biology amidst noise

Separating biological information from experimental noise in cellular images

The meta-problem

- Drug discovery is expensive and time-consuming
- Carrying out meaningful biological experiments is difficult due to experimental noise
- Machine-learning offers the chance to identify biological information faster than humans

Recursion

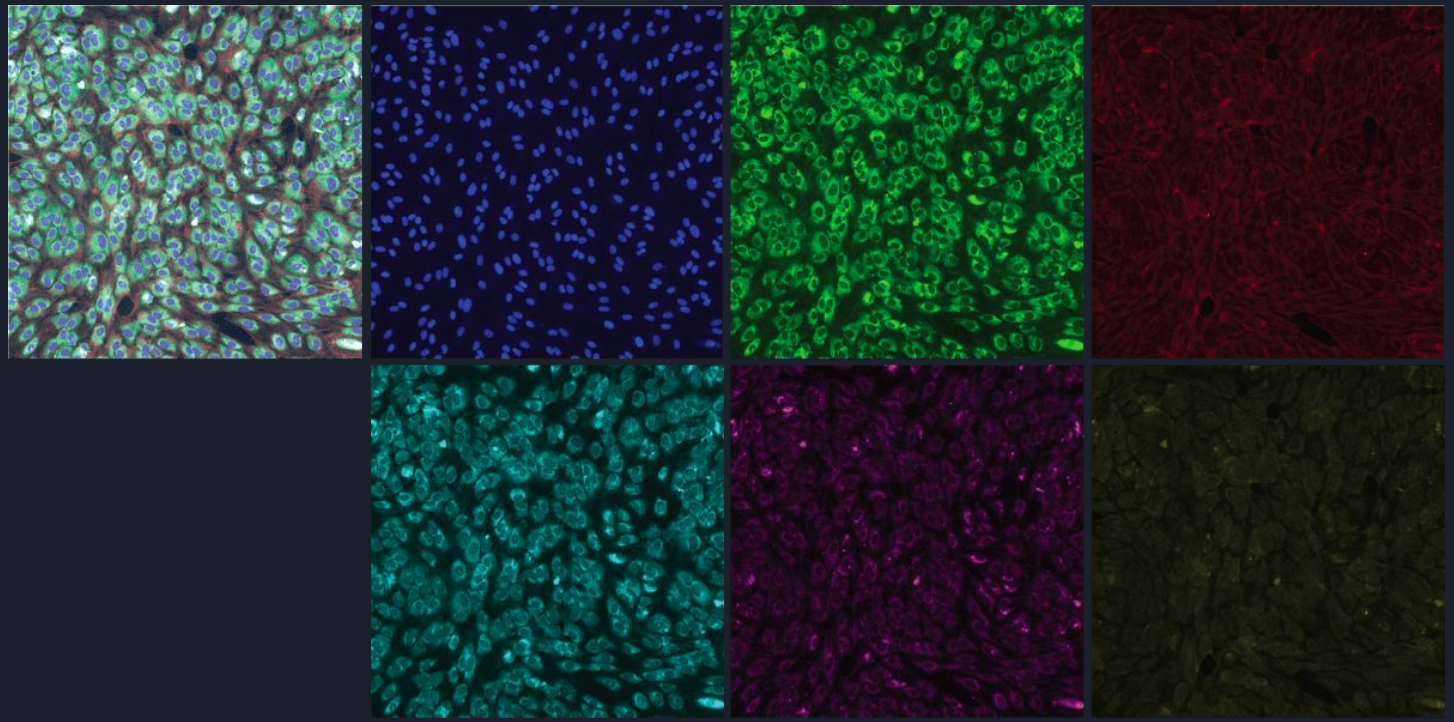


- Established in 2013, Recursion is a “digital biology” company trying to combat these issues
- They bring together experts in machine learning and biology
- Rather than trying to solve a well defined problem, they are target-agnostic

RxRx1

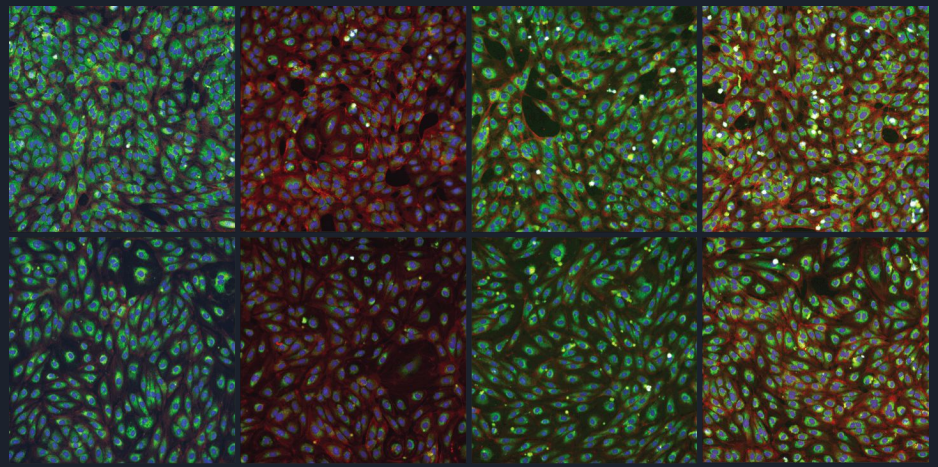
- RxRx1 is Recursion's first publicly available dataset
- 125,510 images with relevant biological and experimental metadata
  - Cell-type: HUVEC, RPE, HepG2, U2OS
  - Plate: one of four
  - Well: one of 308
  - siRNA: 1,108 different small interfering RNAs
- Each image is 512 x 512 x 6 channels

Example image in RxRx1



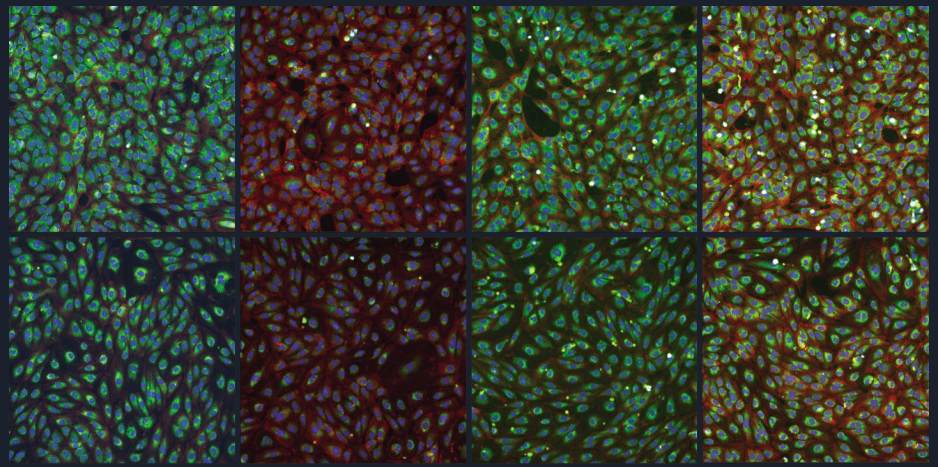
Our task

- Isolate biological information from experimental noise
- Correctly classify the perturbation (siRNA) present in an unseen experiment
- This requires a good classifier, which picks out siRNA independent of batch effects



Potential

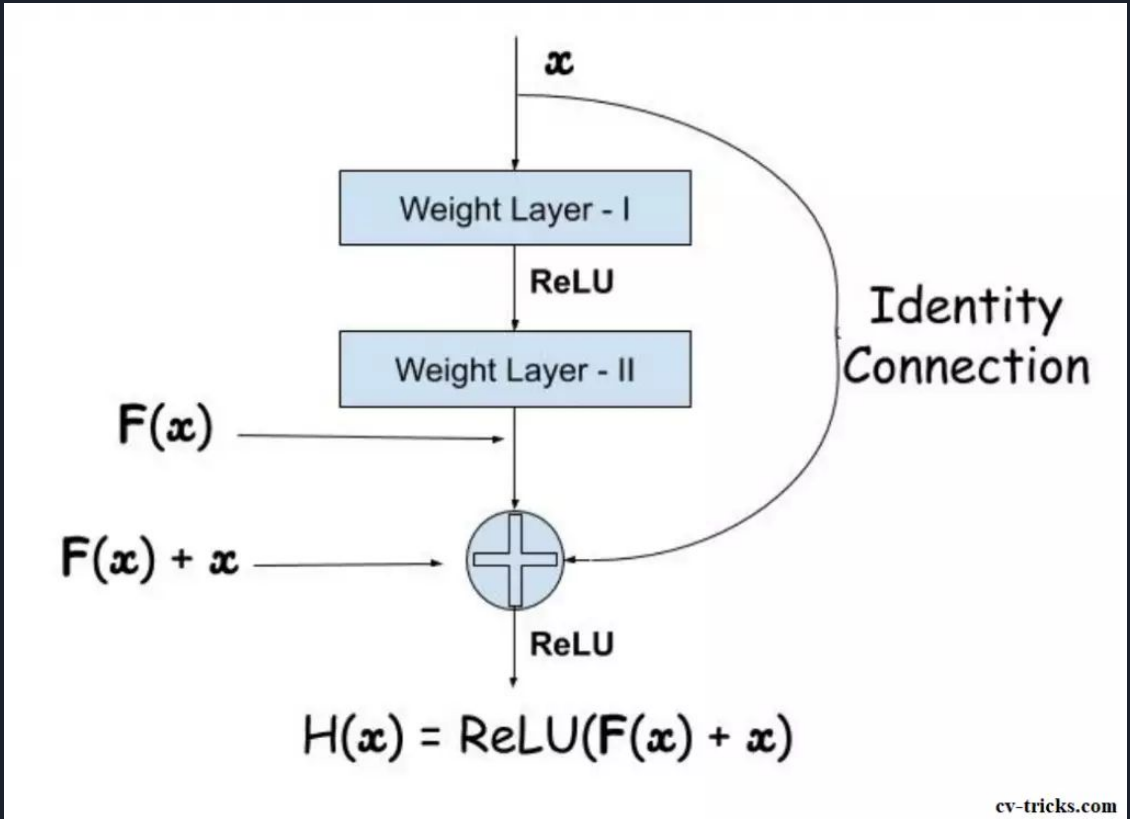
- Isolate biological information from experimental noise
- Correctly classify the perturbation (siRNA) present in an unseen experiment
- This requires a good classifier, which picks out siRNA independent of batch effects



ResNet

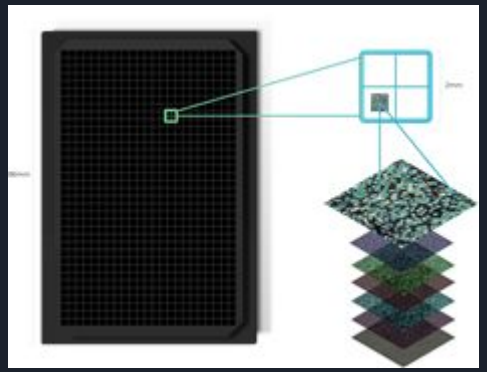
layer name	output size	18-layer	34-layer	50-layer	101-layer	152-layer
conv1	112×112	7×7, 64, stride 2				
conv2_x	56×56	3×3 max pool, stride 2				
		$\begin{bmatrix} 3\times3, 64 \\ 3\times3, 64 \end{bmatrix}\times2$	$\begin{bmatrix} 3\times3, 64 \\ 3\times3, 64 \end{bmatrix}\times3$	$\begin{bmatrix} 1\times1, 64 \\ 3\times3, 64 \\ 1\times1, 256 \end{bmatrix}\times3$	$\begin{bmatrix} 1\times1, 64 \\ 3\times3, 64 \\ 1\times1, 256 \end{bmatrix}\times3$	$\begin{bmatrix} 1\times1, 64 \\ 3\times3, 64 \\ 1\times1, 256 \end{bmatrix}\times3$
conv3_x	28×28	$\begin{bmatrix} 3\times3, 128 \\ 3\times3, 128 \end{bmatrix}\times2$	$\begin{bmatrix} 3\times3, 128 \\ 3\times3, 128 \end{bmatrix}\times4$	$\begin{bmatrix} 1\times1, 128 \\ 3\times3, 128 \\ 1\times1, 512 \end{bmatrix}\times4$	$\begin{bmatrix} 1\times1, 128 \\ 3\times3, 128 \\ 1\times1, 512 \end{bmatrix}\times4$	$\begin{bmatrix} 1\times1, 128 \\ 3\times3, 128 \\ 1\times1, 512 \end{bmatrix}\times8$
conv4_x	14×14	$\begin{bmatrix} 3\times3, 256 \\ 3\times3, 256 \end{bmatrix}\times2$	$\begin{bmatrix} 3\times3, 256 \\ 3\times3, 256 \end{bmatrix}\times6$	$\begin{bmatrix} 1\times1, 256 \\ 3\times3, 256 \\ 1\times1, 1024 \end{bmatrix}\times6$	$\begin{bmatrix} 1\times1, 256 \\ 3\times3, 256 \\ 1\times1, 1024 \end{bmatrix}\times23$	$\begin{bmatrix} 1\times1, 256 \\ 3\times3, 256 \\ 1\times1, 1024 \end{bmatrix}\times36$
conv5_x	7×7	$\begin{bmatrix} 3\times3, 512 \\ 3\times3, 512 \end{bmatrix}\times2$	$\begin{bmatrix} 3\times3, 512 \\ 3\times3, 512 \end{bmatrix}\times3$	$\begin{bmatrix} 1\times1, 512 \\ 3\times3, 512 \\ 1\times1, 2048 \end{bmatrix}\times3$	$\begin{bmatrix} 1\times1, 512 \\ 3\times3, 512 \\ 1\times1, 2048 \end{bmatrix}\times3$	$\begin{bmatrix} 1\times1, 512 \\ 3\times3, 512 \\ 1\times1, 2048 \end{bmatrix}\times3$
	1×1	average pool, 1000-d fc, softmax				
FLOPs		1.8×10^9	3.6×10^9	3.8×10^9	7.6×10^9	11.3×10^9

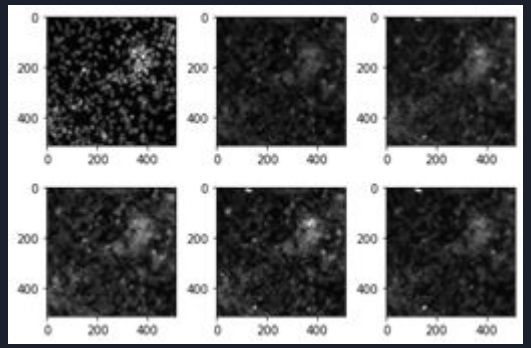
ResNet


x
Weight Layer - I
ReLU
Weight Layer - II
F(x)
F(x) + x
ReLU
Identity Connection
H(x) = ReLU(F(x) + x)
cv-tricks.com

Practical side of the dataset:

Many images: 125k; **Many labels**: ~1k; **6 channels**; **pairs of input** with the same label





Ideas how to train with 6 Channels:

- make a model that takes as the input images 512x512x6

- use pretrained models – then we should(?) have RGB images

    * drop some channels

    * combine them to 512x512x3 during preprocessing

    * add a layer that learns how to combine them to 512x512x3

Ideas to use 2 images with the same output:

- take average/multiply/smth else of predictions from the end layer / multiply / other function like this

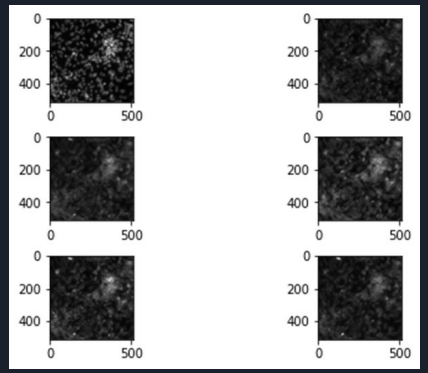
[RGB - Stands for "Red Green Blue." RGB refers to three hues of light that can be mixed together to create different colors]

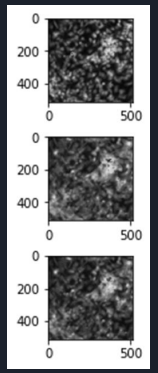
The Model

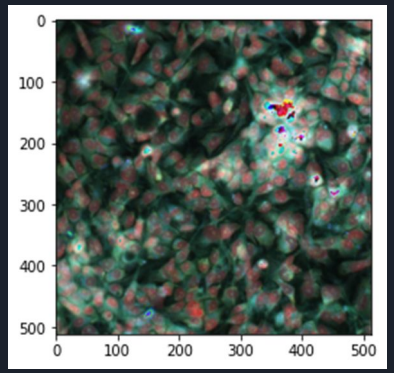
- Feature extraction – update only the last layer and use weights of a pretrained model

  ResNet18 using tutorial: https://pytorch.org/tutorials/beginner/finetuning_torchvision_models_tutorial.html

- Training with 4k images from 125k available
- Merging pairs of channels to treat the image as RGB

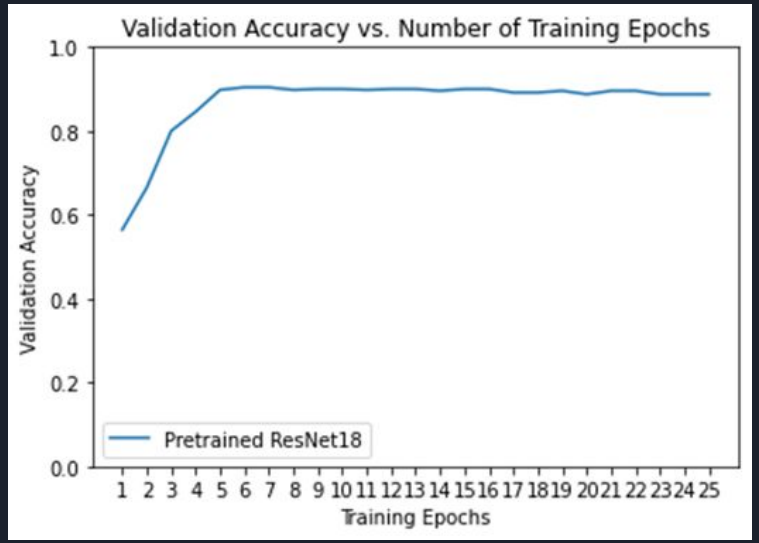






- Not using the fact that there are 2 images from the same site

Results




```
----------
train Loss: 0.0268 Acc: 1.0000
val Loss: 0.7930 Acc: 0.8958
Epoch 21/24
----------
train Loss: 0.0260 Acc: 1.0000
val Loss: 0.7952 Acc: 0.8958
Epoch 22/24
----------
train Loss: 0.0246 Acc: 1.0000
val Loss: 0.7985 Acc: 0.8875
Epoch 23/24
----------
train Loss: 0.0226 Acc: 1.0000
val Loss: 0.7969 Acc: 0.8875
Epoch 24/24
----------
train Loss: 0.0201 Acc: 1.0000
val Loss: 0.7931 Acc: 0.8875
Training complete in 194m 0s
Best val Acc: 0.904167
```

```
best accuracy = Epoch 5/24 with val Loss: 1.0214 Acc: 0.9042
```

Learns fast but overfits very easily - even when used with bigger dataset

[Overfitting - a modeling error that occurs when a function is too closely fit to a limited set of data points]

Further ideas:

- Use more images
    - in each batch choose randomly 5k images from the 125k dataset
- Use images from the same well (with the same output)
- Make better use of 6 channels
    - experiment with different approaches
- Try other models
    - do feature-extraction on pre trained Squeezenet, AlexNet...
    -write some easy, custom CNN

Alternative approach to the dataset

We have a massive dataset of different genetic perturbations.

Questions:

- What are the common features of the dataset?
- What are the differences in features between different genetic perturbations?

…. We can use autoencoders to explore this!

Autoencoders: The Main Idea

- Unsupervised learning algorithm
- Model learns a compressed representation of a set of inputs and its reconstruction
- Can be used for feature extraction and as a generative model

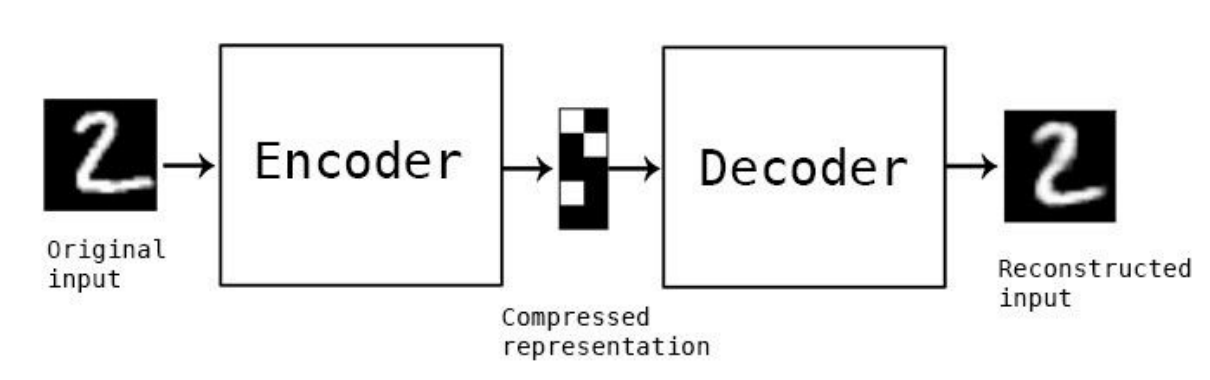


Image Credits: https://towardsdatascience.com/auto-encoder-what-is-it-and-what-is-it-used-for-part-1-3e5c6f017726

Autoencoders: The Main Idea

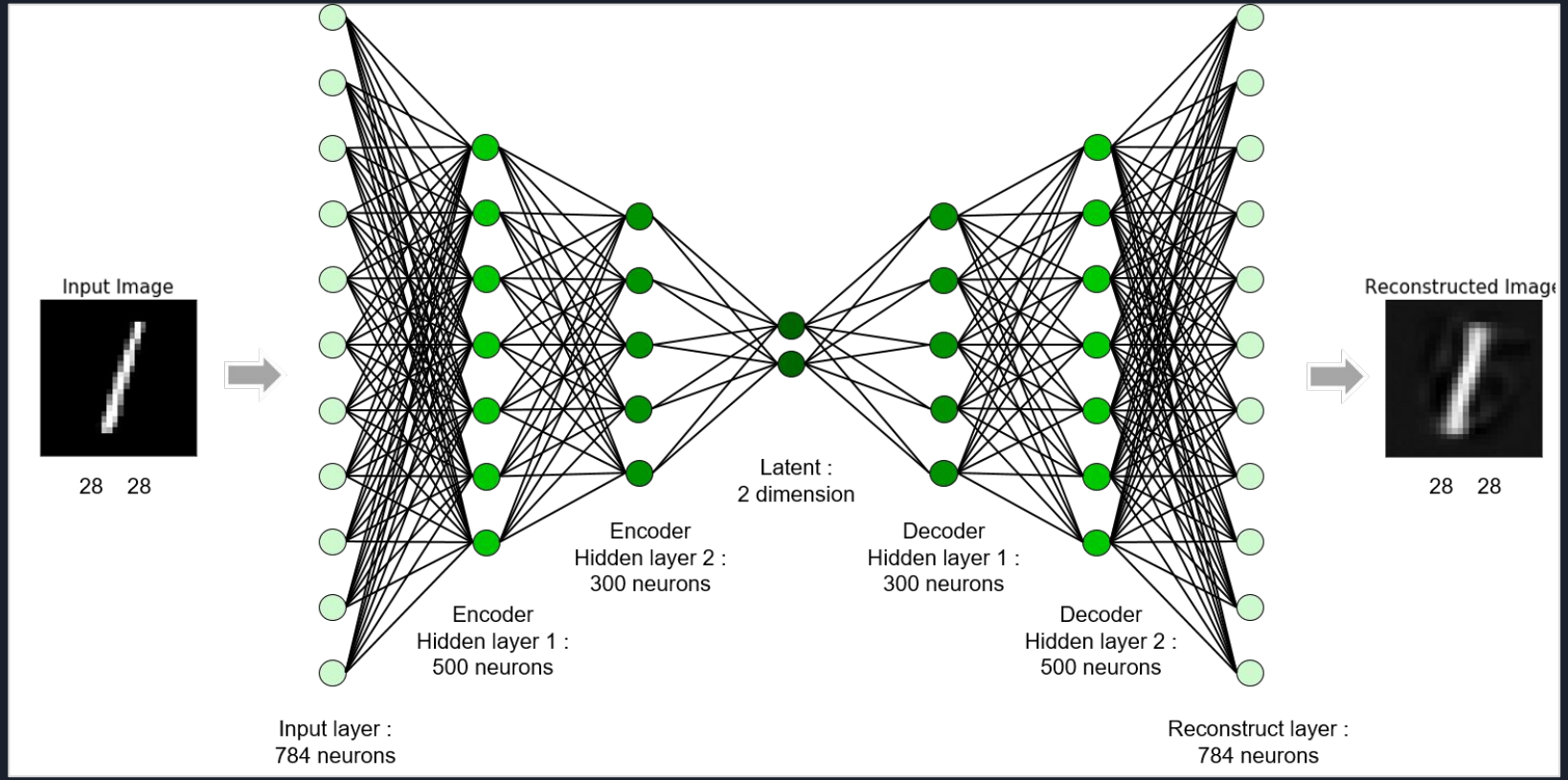


Image Source: https://medium.com/@encodebox/auto-encoder-in-biology-9264da118b83

How Autoencoders can be used on our dataset

Answering “What are the main features of the images? “

- Extracting features from images of each genetic perturbation.
- Comparing features extracted from different genetic perturbations.

The aim is to gain insight on how different genetic perturbations present.

Thank you!

Any Questions?